origin ($t = 0$). Here the a、b spring midpoint is fixed. a、b form two fixed-end spring oscillators: natural length $l/2$, stiffness $2k_1$. Likewise c、d form two oscillators, each of natural length $l/2$ and stiffness $2k_2$.

For a、b、c、d, x_a'、x_b'、x_c'、x_d' are displacements from their own equilibria. Then

$$x_a' = A_1\cos(\omega_1 t + \varphi_1)$$
$$x_b' = A_1\cos(\omega_1 t + \varphi_2)$$

where $\omega_1 = \sqrt{\dfrac{2k_1}{m}}$

A_1、φ_1、φ_2 are constants to determine.

$$x_c' = A_2\cos(\omega_2 t + \varphi_3)$$
$$x_d' = A_2\cos(\omega_2 t + \varphi_4)$$

where $\omega_2 = \sqrt{\dfrac{2k_2}{m}}$

A_2、φ_3、φ_4 are constants to determine.

At $t = 0$, relative to the center-of-mass frame,

$$x_{a0}' = 0 \qquad v_{a0}' = \frac{v_0}{2}$$
$$x_{b0}' = 0 \qquad v_{b0}' = -\frac{v_0}{2}$$
$$x_{c0}' = 0 \qquad v_{c0}' = \frac{v_0}{2}$$
$$x_{d0}' = 0 \qquad v_{d0}' = -\frac{v_0}{2}$$

These are the initial conditions.

Reference circles give four center-frame velocities, respectively:

$$v_a' = -A_1\omega_1\sin(\omega_1 t + \varphi_1)$$
$$v_b' = -A_1\omega_1\sin(\omega_1 t + \varphi_2)$$
$$v_c' = -A_2\omega_2\sin(\omega_2 t + \varphi_3)$$
$$v_d' = -A_2\omega_2\sin(\omega_2 t + \varphi_4)$$

Displacements, velocities and initial conditions give

$$A_1 = \frac{v_0}{2\omega_1} \qquad \varphi_1 = -\frac{\pi}{2} \qquad \varphi_2 = \frac{\pi}{2}$$
$$A_2 = \frac{v_0}{2\omega_2} \qquad \varphi_3 = -\frac{\pi}{2} \qquad \varphi_4 = \frac{\pi}{2}$$

Hence

$$\left.\begin{aligned} x_a' &= \frac{v_0}{2\omega_1}\sin\omega_1 t \\ x_b' &= -\frac{v_0}{2\omega_1}\sin\omega_1 t \\ x_c' &= \frac{v_0}{2\omega_2}\sin\omega_2 t \\ x_d' &= -\frac{v_0}{2\omega_2}\sin\omega_2 t \end{aligned}\right\} \qquad (1)$$